In demo 2 (constraint-aware)

- agent queries constraints
- residual indicates the action will fail
- agent avoids the attempt entirely and informs the user

Because the agent integrates policy-derived constraints into its reasoning process, failed tool invocations are rare.

This demo illustrates the constraint-aware planning model:

- the agent queries policy before acting
- the PDP returns policy residuals that describe remaining constraints
- the agent plans within those constraints

This reduces trial-and-error behavior and improves efficiency while maintaining strong enforcement guarantees.

C.5 Demo 3: delegation as data

The third demonstration introduces delegation. This allows a main agent to grant a subagent a limited subset of its authority. At a high level, the flow is:

- the main agent creates a delegation record
- the subagent attempts actions
- the PEP creates a request that includes the delegation data
- PDP evaluates the request with delegation context

To test that the policies are working, run the delegation policy validation script:

```
python3 demo/test-delegation.py
```

A typical output looks like this: